WORKED SOLUTION

STEP I 2006, Question 12

Probability and Statistics • Conditional Probability

STEP I 2006, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/06-s1-q12

The key idea

A road is open only if *every* section on it is open, and the towns are cut off only if *every* road is blocked. So we work road by road, using independence, and always through complements. The second part is a conditional probability in which the choice of road matters: a one-section road is more likely to be open than a two-section road, so finding your road open tells you something about which road you chose. We handle that with the definition $\mathrm{P}(B \mid A) = \mathrm{P}(A \cap B)/\mathrm{P}(A)$, splitting by the road chosen.

Solution

Probability of being cut off

Call the two-section roads R_1 and R_2 and the one-section road R_3. Let $q = 1 - p$ be the probability that a single section is open.

A two-section road is open exactly when both of its sections are open, which by independence has probability $q^2 = (1-p)^2$. So

$$\mathrm{P}(R_1 \text{ blocked}) = \mathrm{P}(R_2 \text{ blocked}) = 1 - (1-p)^2 = p(2-p),$$
$$\mathrm{P}(R_3 \text{ blocked}) = p.$$

The three roads use five different sections, so the events "R_1 blocked", "R_2 blocked" and "R_3 blocked" depend on disjoint sets of independent sections and are therefore independent. Oxtown is cut off from Camville exactly when all three roads are blocked, so

$$\mathrm{P}(\text{cut off}) = p(2-p) \cdot p(2-p) \cdot p = p^3(2-p)^2,$$

as required.

The conditional probability

Let C_k be the event that I choose road R_k, so $\mathrm{P}(C_k) = \frac{1}{3}$ for each k, and my choice is independent of the flooding. Let O be the event that my road is open, and B the event that the other two roads are both blocked. We want $\mathrm{P}(B \mid O) = \mathrm{P}(B \cap O)/\mathrm{P}(O)$.

> **Ask yourself** Isn't the answer just P(the other two roads blocked), by independence?

It would be if we knew which road we had chosen. But "the other two roads" are a different pair depending on our choice, and the observation that our road is open makes the short road R_3 a more likely choice than before. So we must split by the road chosen and weight correctly, which is exactly what the formula below does.

The denominator. Using the law of total probability over the choice of road,

$$\begin{aligned}\mathrm{P}(O) &= \tfrac{1}{3}(1-p)^2 + \tfrac{1}{3}(1-p)^2 + \tfrac{1}{3}(1-p) \\ &= \tfrac{1}{3}(1-p)[2(1-p)+1] = \tfrac{1}{3}(1-p)(3-2p).\end{aligned}$$

The numerator. If I chose R_1, I need R_1 open and R_2, R_3 blocked; by independence of the roads this has probability $(1-p)^2 \cdot p(2-p) \cdot p$. Choosing R_2 gives the same. If I chose R_3, I need R_3 open and both two-section roads blocked: $(1-p) \cdot p^2(2-p)^2$. Hence

$$\begin{aligned}\mathrm{P}(B \cap O) &= \tfrac{1}{3}\left[2p^2(2-p)(1-p)^2 + p^2(2-p)^2(1-p)\right] \\ &= \tfrac{1}{3}\,p^2(1-p)(2-p)[2(1-p)+(2-p)] \\ &= \tfrac{1}{3}\,p^2(1-p)(2-p)(4-3p).\end{aligned}$$

The answer. Dividing, and cancelling the factor $\frac{1}{3}(1-p)$, which is non-zero for $p < 1$,

$$\mathrm{P}(B \mid O) = \frac{p^2(2-p)(4-3p)}{3-2p}.$$

Each of the factors p, $2-p$, $4-3p$ and $3-2p$ is linear, so this is fully factorised.

The limit $p \to 1$

As $p \to 1$ every factor tends to 1, so

$$\mathrm{P}(B \mid O) \to \frac{1 \cdot 1 \cdot 1}{1} = 1.$$

This makes sense. When flooding is almost certain, finding my own road open is a lucky fluke, and the other roads, which flood independently of mine, almost certainly did not share my luck: the knowledge that my road is open says nothing about theirs being open. So I should be nearly certain that any other choice would have failed. (At $p = 1$ itself the conditioning event has probability zero and the conditional probability is undefined; only the limit is meaningful.)

Checks

- At $p = 0$ nothing is ever blocked, and the formula gives 0, as it should.
- At $p = \frac{1}{2}$ the formula gives $\dfrac{\frac{1}{4} \cdot \frac{3}{2} \cdot \frac{5}{2}}{2} = \frac{15}{32}$. Enumerating all 2^5 patterns of blocked sections and all three road choices exactly gives the same value, and likewise for several other values of p, such as $p = \frac{2}{3}$, giving $\frac{32}{45}$.
- The answer is always between 0 and 1 for $0 \leq p < 1$: for example at $p = 0.99$ it is about 0.9996.

Common slips

- Taking a two-section road to be blocked with probability p^2 (both sections blocked) rather than $1 - (1 - p)^2$ (at least one blocked).
- Forgetting to condition: the probability of the other two roads being blocked is not the same as its conditional probability given that my randomly chosen road is open.
- Weighting the three choices equally *after* the observation; it is the prior probabilities $\frac{1}{3}$ that are equal, not the posterior ones.

Beyond STEP

Roads in parallel, each made of sections in series, is the simplest example of a *reliability network*; the probability that a network connects two points is studied in general in percolation theory, where the sharp change in behaviour as p varies is a central theme. The second part is Bayes' theorem in disguise: the observation "my road is open" updates the probabilities of which road was chosen. Grimmett and Welsh, *Probability: An Introduction*, covers conditional probability and independence in exactly this spirit.

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